AGENTIC AI PROFESSIONAL TRACK

Five Quarters to **Certified Agentic AI Architect**

From mastering AI agents to building and deploying them at cloud scale — earn the Certified Agentic AI Architect credential across four levels.

AI-101 → AI-151 → AI-251 → AI-321 → AI-451

5 Quarters · 4 Certification Levels · 12 Exams

PIAIC and PANAVERSITY
The AI Agent Factory
agentfactory.panaversity.org

Program Overview

The Certified Agentic AI Architect is a five-quarter professional certification program for software practitioners and AI engineers. It takes you from AI-driven development fundamentals through building and deploying Agent Factories at cloud scale.

The program follows a progressive four-level certification structure. Each level is earned by passing the corresponding quarter's exams. All four levels culminate in the Certified Agentic AI Architect credential. Critical thinking skills from Part 0 (Thinking is the Curriculum) and hands-on OpenClaw projects run as parallel tracks throughout.

Certification Ladder

Level	Certification	Quarter(s)	Exams
1	**Certified Agentic AI Foundations**	Q1 + Q2	4 Exams
2	**Certified Agentic AI Developer**	Q3	2 Exams
3	**Certified Agent Factory Builder**	Q4	2 Exams
4	**Certified Agentic AI Engineer**	Q5	2 Exams
★	**All Levels Completed → Certified Agentic AI Architect**		

The Learning Path

Course	Title	Key Domains	Level Earned
AI-101	AI-Driven Development, Claude Code, and OpenClaw Fundamentals	Claude Code, OpenClaw, AI Agents, Markdown Specs	☆ Contributes to Level 1
AI-151	Context Engineering, Spec-Driven Development, and Advanced Claude Code	Context Engineering, Agentic Workflows, Version Control	★ **Earns Level 1: Certified Agentic AI Foundations**
AI-251	Learn to Think and Program in the AI Era	Python, Type-Driven AI, Testing, OOP, CLI & APIs	★ **Earns Level 2: Certified Agentic AI Developer**
AI-321	Building AI Agent Factory	Claude Agent SDK, OpenAI Agents SDK, MCP, FastAPI, RAG	★ **Earns Level 3: Certified Agent Factory Builder**
AI-451	Deploying Agent Factories in the Cloud	Docker, Kubernetes, Kafka, CI/CD, Multi-Cloud	★ **Earns Level 4: Certified Agentic AI Engineer**

Curriculum Details

- Quarter 1 (Level 1)

AI-101

AI-Driven Development, Claude Code, and OpenClaw Fundamentals

Your entry point into the Agent Factory paradigm. Learn why AI is non-negotiable, master Claude Code as your primary development tool, build your first OpenClaw applications, and begin developing critical thinking skills that separate you from AI-dependent practitioners.

Book Coverage: Preface + About + Thesis + Part 1: General Agents Foundations (Chapters 12–14) + Part 5: Building OpenClaw Apps + Part 0: Chapters 1–2

What You Will Learn

- The Agent Factory Thesis — the paradigm shift from writing software to manufacturing AI employees
- Which AI Employees Should You Use in 2026 — navigating the agent landscape
- Why AI Is Non-Negotiable — the economic imperative for every organisation
- The Agent Factory Paradigm — Digital FTEs, the Two-Layer Model, and the 10-80-10 Rule
- Markdown Writing Instructions — the foundational skill for spec-driven development
- General Agents — understanding agent architecture and capabilities
- OpenClaw Fundamentals — meet your first AI employee and build your first apps
- Asking Better Questions — question formulation that AI cannot replicate (Part 0)
- Detecting Broken Reasoning — spotting errors in AI-generated outputs (Part 0)

Assessments

- ☐ **Mid-Term: Agent Factory Intro Exam** (Exam L1: P1-AFI)
- ☐ **Final: Claude Code Foundation Exam** (Exam L1: P2-CCF)

Parallel Tracks

- OpenClaw Fundamentals — Online Classes and Hackathon
- Learn to Think Part 1 — Online Classes and Projects (Chapters 1–2)

Key Skills:

Claude Code · OpenClaw · AI Agents · Markdown Specs · Critical Thinking · Agent Factory

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☆ **Contributes to: Certified Agentic AI Foundations (contributes 2 of 4 exams)**

- Quarter 2 (Level 1)

AI-151

Context Engineering, Spec-Driven Development, and Advanced Claude Code

Master the art of context engineering and spec-driven development — the methodology that makes AI agents reliable. Build agent workflow primitives for file processing, structured data, and version control. Complete Level 1 certification.

Book Coverage: Part 1: General Agents Foundations (Chapters 15–18) + Part 2: Agent Workflow Primitives + Part 5: OpenClaw Projects (Chapters 58–60) + Part 0: Chapters 3–5

What You Will Learn

- Context Engineering — the discipline of providing AI with exactly the right information
- Spec-Driven Development — writing specifications that agents can execute reliably
- The Seven Principles — foundational laws for building production-grade agent systems
- Claude Code for Teams & CI/CD — collaborative development and continuous integration
- File Processing Workflows — building agents that handle document pipelines
- Structured Data & Persistent Storage — agents that work with databases and state
- Version Control Workflows — agents that manage code repositories intelligently
- Thinking in Systems — seeing second-order consequences (Part 0)
- Reasoning From First Principles — knowing what to automate and what not to (Part 0)
- Communicating What Matters — stakeholder communication under pressure (Part 0)

Assessments

- ☐ **Mid-Term: Context Engineering & Advanced Claude Code** (Exam L1: P3-ACC)
- ☐ **Final: Agent Workflow Primitives** (Exam L1: P4-AWP)

Parallel Tracks

- OpenClaw Project — Online Classes and Hackathon (Chapters 58–60)
- Learn to Think Part 2 — Online Classes and Projects (Chapters 3–5)

Key Skills:

Context Engineering · Spec-Driven Dev · Agentic Workflows · File Processing · Structured Data · Version Control

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★ **Earns: Certified Agentic AI Foundations**

- Quarter 3 (Level 2)

AI-251

Learn to Think and Program in the AI Era

The programming quarter. Eight phases take you from setting up the development workbench through type-driven development, testing, debugging, OOP, real-world Python, CLI applications, and production systems. In parallel, complete the remaining Thinking is the Curriculum chapters and assemble your Thinking Portfolio.

Book Coverage: Part 4: Programming in the AI Era (Chapters 34–54, 8 Phases) + Part 0: Chapters 6–11

What You Will Learn

- Phase 1: The Workbench — setting up the professional development environment (Chapters 42–46)
- Phase 2: Specify with Types — types as the language of intent (Chapters 34–37)
- Phase 3: Tests as Specification — test-driven verification (Chapters 38–41)
- Phase 4: Debug & Master — debugging skills and TDG independence (Chapters 42–43)
- Phase 5: The Python Object Model — OOP done right (Chapters 44–47)
- Phase 6: Real-World Python — building practical applications (Chapters 48–50)
- Phase 7: CLI & Concurrency — command-line tools and parallel execution (Chapters 51–52)
- Phase 8: Production Systems — hardening and securing code (Chapters 53–54)
- Working With AI, Not For AI — the AI collaboration framework (Part 0)
- Reasoning Through Dilemmas, Building From Nothing, Deciding Under Uncertainty (Part 0)
- Learning How to Learn + Thinking Portfolio assembly (Part 0, Chapters 10–11)

Assessments

- ☐ **Mid-Term: Fundamentals of Programming in the AI Era** `(Exam L2: P1-FPA)`
- ☐ **Final: Advanced AI Programming in the AI Era** `(Exam L2: P2-APA)`

Parallel Tracks

- Learn to Think Part 3 — Online Classes and Projects (Chapters 6–11, completing Part 0)

Key Skills:

`Python · Type-Driven AI · Testing · OOP · CLI & APIs · Production Systems · Debugging`

★ Earns: Certified Agentic AI Developer

- Quarter 4 (Level 3)

AI-321

Building AI Agent Factory

The construction quarter. Build production-grade Agent Factories using the Claude Agent SDK and OpenAI Agents SDK. Master MCP server development, FastAPI integration, and RAG pipelines. Every chapter produces a deployable agent system.

Book Coverage: Part 6: Building Agent Factories (complete)

What You Will Learn

- Claude Agent SDK — the primary framework for building custom agents
- OpenAI Agents SDK — comparative agent development framework
- Claude API — direct API integration for custom workflows
- Model Context Protocol (MCP) — building MCP servers and connectors
- FastAPI — production API endpoints for agent systems
- Retrieval-Augmented Generation (RAG) — grounding agents in organisational knowledge
- Agent Factory Architecture — the Factory Layer + Edge Layer pattern
- Multi-agent orchestration and handoff patterns

Assessments

- ☐ **Mid-Term: Building Agent Factories Fundamentals** (Exam L3: P1-BAFF)
- ☐ **Final: Building Advanced Agent Factories** (Exam L3: P2-BAAF)

Key Skills:

Claude Agent SDK · OpenAI Agents SDK · Claude API · MCP · FastAPI · RAG · Multi-Agent

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★ **Earns: Certified Agent Factory Builder**

- Quarter 5 (Level 4)

AI-451

Deploying Agent Factories in the Cloud

The deployment quarter. Take your Agent Factories from development to production at cloud scale. Master containerisation, orchestration, event streaming, continuous deployment, and multi-cloud strategies.

Book Coverage: Part 7: Deploying Agent Factories in the Cloud (complete)

What You Will Learn

- Docker — containerising agent systems for consistent deployment
- Kubernetes — orchestrating agent workloads at scale
- Kafka — event streaming for real-time agent communication
- CI/CD Pipelines — automated testing, building, and deployment
- Multi-Cloud Deployment — strategies for AWS, GCP, and Azure
- Production Monitoring — observability, logging, and alerting for agent systems
- Security & Compliance — securing agent deployments in enterprise environments
- Cost Optimisation — managing cloud spend for agent infrastructure

Assessments

- ☐ **Mid-Term: Agentic AI Cloud Deployment Fundamentals** (Exam L4: P1-AACDF)
- ☐ **Final: Advanced Agentic AI Cloud Deployment** (Exam L4: P2-AAACD)

Key Skills:

Docker · Kubernetes · Kafka · CI/CD · Multi-Cloud · Monitoring · Security

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★ **Earns: Certified Agentic AI Engineer**

> ALL FOUR LEVELS COMPLETED
>
> ## ★ Certified Agentic AI Architect
>
> The capstone credential. Awarded upon completion of all four certification levels (10 exams) across 5 quarters. Certifies end-to-end mastery: from AI-driven development and critical thinking through agent construction to cloud-scale deployment.

Two AI Tracks, One Ecosystem

Panaversity offers two parallel certification tracks. Choose the one that matches your background. Both produce professionals who can deploy AI agents at enterprise scale—one through code, the other through domain expertise.

Agentic AI Professional Track (This Program)	Agentic AI Business & Domain Expert Track
Courses: AI-101 → AI-151 → AI-251 → AI-321 → AI-451	**Courses:** BAI-100 → BAI-200 → BAI-300
Duration: 5 Quarters	**Duration:** 3 Quarters
Primary Tool: Claude Code + Claude Agent SDK	**Primary Tool:** Claude Cowork
Core Focus: Code, agents, APIs, cloud deployment	**Core Focus:** Thinking, domain knowledge, enterprise workflows
Builds: Custom agents, Agent Factories, cloud infrastructure	**Builds:** SKILL.md libraries, Cowork plugins, Agentic Office
Final Certification: Certified Agentic AI Architect	**Final Certification:** Certified Agentic AI Business Strategist
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